Day-Ahead Energy Consumption Forecasting

A Share-Based Approach for Iberia Retail

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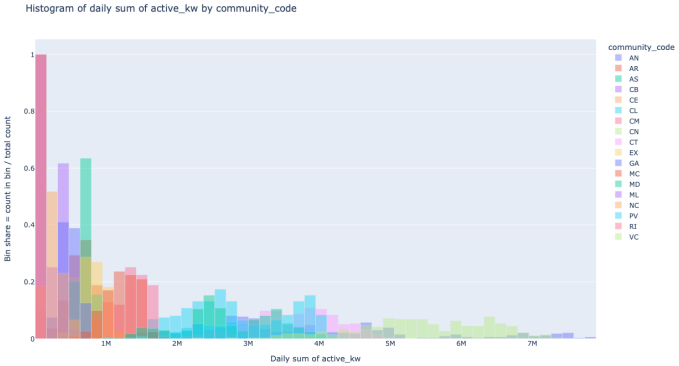
Iberia Retail Portfolio — 15-minute MAE minimisation

Roadmap

1. Why Modelling `active_kw` Directly is Hard
2. The Share-Based Formulation
3. Baseline: α-Lag-7
4. Closing the Gap — Residual Model
5. What We Tried & Final Submission

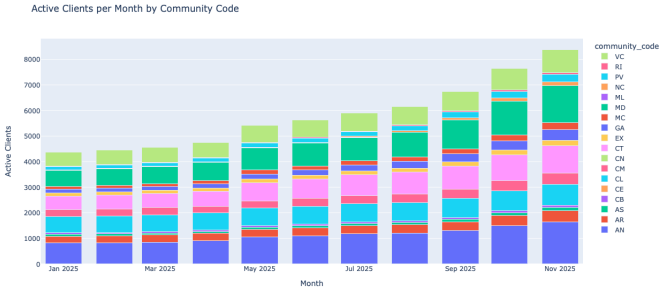
Heterogeneous Consumption Across Communities

- Daily $\sum$`active_kw` varies by more than an order of magnitude across the 17 communities
- A single model on raw `active_kw` would be dominated by the largest ones
- Within a community, consumption depends on calendar, weather, working hours — very noisy in absolute units



Daily $\sum$ `active_kw` per community — widely different scales and shapes

A Growing Portfolio — Learn Behaviour, Not Growth



Active clients per month, stacked by community

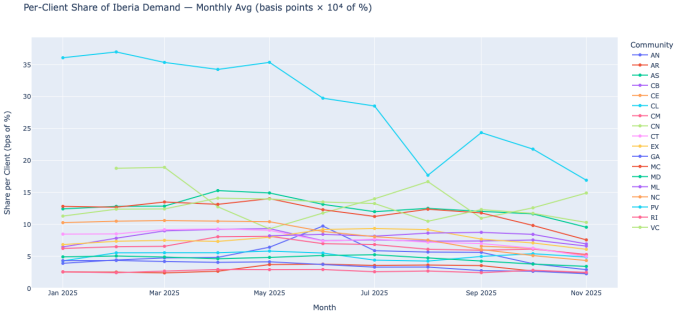
- Portfolio **roughly doubled** over 2025 ($\approx$4.4k $\rightarrow$ $\approx$8.4k clients)
- A model on total `active_kw` would absorb this trend and extrapolate it blindly
- Fix: normalise by **# active clients** and by **Spain demand forecast** $\hat{D}(t)$ — the forecast already encodes calendar, seasonality, weather

The α Target — Per-Community Per-Client Share of Demand

$$\alpha(c,t) \;=\; \frac{\sum_{j\in c} y_j(t)}{n(c,t)\;\cdot\;\hat{D}_{\text{Spain}}(t)}$$

y_j: client active_kw $\quad n(c,t)$: active clients in community c $\quad \hat{D}_{\text{Spain}}$: REE demand forecast

- Denominator $\hat{D}$ absorbs calendar, temperature, daily & weekly seasonality
- Division by $n(c,t)$ neutralises portfolio growth
- Most communities sit in a narrow band $\Rightarrow \alpha$ is near-stationary



Baseline — Last Week's α is Already a Strong Predictor

To predict the target $\alpha(c,t)$ we simply reuse its value from a week ago:

$$\hat{\alpha}(c,t) \;=\; \alpha(c,\, D{-}7)$$

$$\hat{y}_{\text{portfolio}}(t) \;=\; \sum_{c} \underbrace{\alpha(c,\, D{-}7)}_{\text{observed}} \cdot \underbrace{n(c,\, D{-}2)}_{\text{last known}} \cdot \underbrace{\hat{D}_{\text{Spain}}(t)}_{\text{REE forecast}}$$

- Every term is available strictly before 12:00 on $D{-}1$ — no leakage
- Strong weekly periodicity of human activity $\Rightarrow$ $\alpha(c,\, D{-}7) \approx \alpha(c,t)$
- Multiplying back by today's $\hat{D}(t)$ (forecasted) re-injects today's weather and calendar
- Pure structural model — zero fitted parameters

ZERO fitted parameters

ZERO hyperparameters

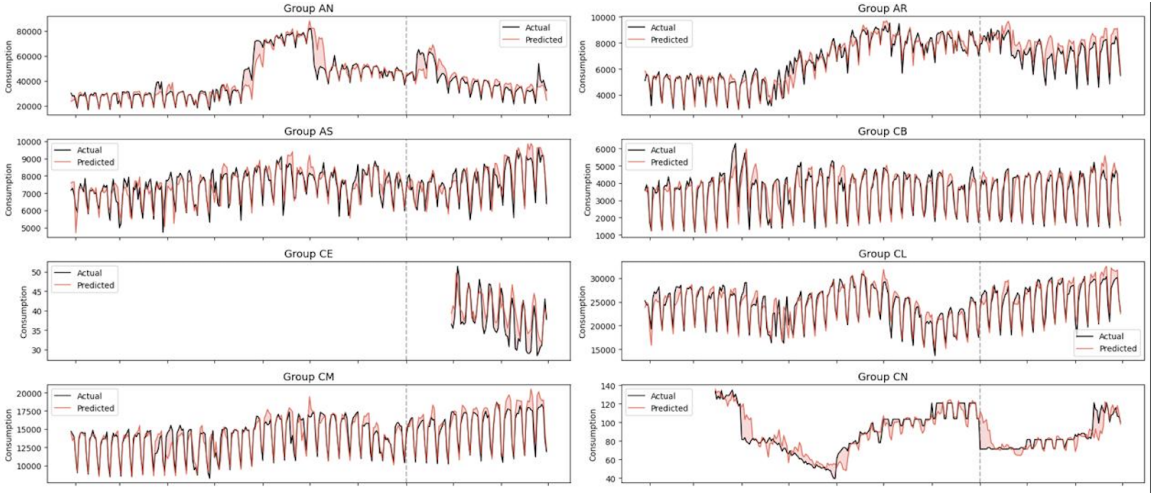
Residual Model — A Natural Next Step

$$\varepsilon(c,t) \;=\; \sum_{j\in c} y_j(t) \;-\; \alpha(c,\, t-7\mathrm{d}) \cdot n(c,\, D{-}2) \cdot \hat{D}_{\mathrm{Spain}}(t)$$

- From here, the obvious next step is to **fit the residual ε per community** with a learned model
- Many external signals the baseline ignores could help:
  - Wind and PV production forecasts
  - Weather — temperature, HDD/CDD, radiation, humidity, wind speed
  - Public & regional holidays, bridge days
  - Time-of-day, day-of-week, day-of-month, day-of-year encodings
- The structural baseline already absorbs most of the signal, so ε should be smaller, closer to zero-mean, and in principle easier to fit

Solution

- Fitted a **LightGBM per community** on the residual $\varepsilon(c, t)$
- Features fed to each model:
  - Weather (forecasted D-2) — temperature, HDD/CDD, radiation, humidity, wind speed
  - Time / day / month encodings — 15-min interval, day-of-week, day-of-month, day-of-year, sin / cos cyclical features
- Final portfolio forecast sums the per-community predictions on top of the α-lag-7 baseline


Group AN
Group AR
Group AS
Group CB
Group CE
Group CL
Group CM
Group CN
Actual
Predicted
Consumption

Thank you

Arthur Vianna **Yanis Fallet** **Nicolaj Thomsen** **Arthur Windels**

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Iberia Retail Portfolio at 15-Minute Resolution

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